Hippocampus unfolding
Final experimental report

MSD Hippocampus / SwinUNETR / fold 0 / 7 October 2026

The auxiliary branch helps, but this study does not establish an advantage specific to unwrapping. Its main benefit is fewer false positives. Consistent recovery of missing tissue remains unresolved.

Question	5% data: 10 training cases	Full data: 208 training cases
Does unfolding itself help?	**No clear benefit over the matched plane branch.** Three-seed means: unwrapped 79.73%, plane 80.03% (-0.30 pp).	**No consistent advantage.** Unwrapped 88.21%, plane 88.19% (+0.02 pp); the direction reverses in seed 2. This is not an equivalence test.
Is depth balancing still useful?	Small, architecture-dependent seed-0 effect: +0.21 pp for unwrapped, -0.19 pp for plane. The branch gain remains with Dice alone.	Small seed-0 effect: +0.15 pp for unwrapped, -0.10 pp for plane. Both full-data case-bootstrap intervals include zero.
Are improvements repeatable?	Both branches beat native + depth in all three seeds. Mean gains: plane +5.15 pp; unwrapped +4.85 pp. Seed variability remains large.	Both branches beat native + depth in all three seeds. Mean gains: plane +0.32 pp; unwrapped +0.34 pp.
Does uncertainty locate useful corrections?	Spatial gating beats uniform gating, but **neither beats ordinary fusion**. Unwrapped: 81.47% spatial, 79.72% uniform, 81.89% ordinary.	Same ranking for unwrapped: 88.21% spatial, 87.88% uniform, 88.38% ordinary. Plane spatial and uniform gates are nearly tied, both below ordinary.
Are anatomical errors improving?	Compared with native + depth, unwrapped has far fewer extra voxels and better surfaces, but **more missed tissue in every seed**. A/P swaps are not consistently improved.	Extra voxels and surface distances improve; missed tissue increases in every seed versus native + depth. Swaps improve in two seeds and worsen in one.

Completion: all 24 new runs and four checkpoint audits succeeded. The appendix includes all 20 original and 24 new conditions, with reused runs counted once. All scores use the same 52 validation cases. Dice is reported as a percentage; differences are percentage points (pp).

The strongest inference is repeatable benefit from the overall branch setup. Matched plane controls prevent attributing that benefit to unfolding alone.

What was trained and held fixed

One end-to-end segmentation model

The native SwinUNETR and auxiliary branch were trained together. MRI samples in the auxiliary chart are encoded, mapped back through saved source voxel indices, and fused with native decoder features. Ordinary fusion predicts from **F + R**. No native encoder was frozen.

Approach	Meaning
Native Dice	SwinUNETR with soft multiclass Dice including background; no auxiliary branch.
Native + depth	Same model plus depth-balanced cross-entropy. Signed 2 mm bands have balanced influence within foreground and an 8 mm background margin. Sparse bands are downweighted.
Matched plane branch	Same branch capacity, chart membership and sampled MRI information as unfolding, arranged in plane coordinates. It is the geometric control.
Unwrapped branch	Cut-shell harmonic UV charts with distance from the longitudinal cutting plane. Separate signed sides retain depth; features are mapped into the original segmentation grid.
Local / context / position / attention	Cumulative additions: fine encoder; coarse UV and chart context; longitudinal and signed-depth coordinates; then channel and spatial attention.
Spatial / uniform gates	Multiply the residual R by detached native-path softmax entropy / log(3), either per voxel or averaged over the volume. No added head, parameters or loss.

Geometry and optimization

The plane passes through the foreground centroid and contains the longest physical PCA axis. Its normal is the RAS right axis projected orthogonally to that long axis. Branch inputs use fixed guide predictions, not annotation masks. Annotations define the loss geometry and evaluation strata.

Two out-of-fold 25-epoch guides generate training geometry; a fixed 25-epoch guide supplies validation geometry. Segmentor seeds 0, 1 and 2 share the seed-0 guide atlas and depth coefficients. Replication therefore varies the segmentor conditional on fixed geometry.

64 x 64 x 64 input; feature size 24; batch size 1; AdamW learning rate 0.0001, weight decay 0.00001; StepLR halves the rate every 20 epochs; bfloat16; no augmentation. Maximum 75 epochs; minimum 25; patience 8; min_delta 0.0005. Select the highest mean patient A/P Dice checkpoint.

Depth CE: five Dice-only warm-up epochs, then a five-epoch ramp. Frozen coefficients: 0.0029145569633 at 5%, 0.0803403871895 at full data. No SDF, topology term, or additional anatomical loss was introduced.

Repeatability across three seeds

Data	Model	Seed 0	Seed 1	Seed 2	Mean	SD
5% data	Native + depth	75.12	78.02	71.50	74.88	3.27
5% data	Plane	81.81	82.10	76.18	80.03	3.34
5% data	Unwrapped	81.89	81.76	75.53	79.73	3.63
Full data	Native + depth	88.01	87.73	87.87	87.87	0.14
Full data	Plane	88.19	88.07	88.31	88.19	0.12
Full data	Unwrapped	88.38	88.13	88.13	88.21	0.14

Both branches improve on native + depth in all six matched regime/seed comparisons. The 5% seed-2 runs are lower across all architectures. Unwrapping does not remove this variability.

Unwrapped minus matched plane

Data	Seed	Dice delta (pp)	95% case-bootstrap interval	Cases better / worse
5% data	0	+0.09	[-0.150, +0.323]	26 / 26
5% data	1	-0.34	[-0.658, -0.032]	25 / 27
5% data	2	-0.65	[-0.787, -0.514]	5 / 47
Full data	0	+0.19	[-0.105, +0.485]	30 / 22
Full data	1	+0.06	[-0.307, +0.425]	28 / 24
Full data	2	-0.18	[-0.432, +0.062]	21 / 31

At 5%, the mean unfolding-specific difference is -0.302 pp. At full data it is +0.023 pp. Full-data seed differences change sign, and every per-seed interval includes zero. The small numerical mean does not establish equivalence.

Interpretation of intervals

Intervals resample 52 paired validation cases 10,000 times with RNG seed 0. They condition on already-selected checkpoints, the reused fold and fixed guides. They do not quantify full-pipeline training uncertainty or independent-test performance. SD describes only three segmentor seeds, and no multiple-comparison correction is applied.

The selected 5% pair uses position features; the full-data pair uses attention. The data-regime contrast is not an isolated experiment on training-set size. The original seed-0 plane-attention result (82.06%) also exceeds the selected 5% plane-position result (81.81%).

Depth loss and uncertainty gating

All entries here use seed 0. Gated models retain the depth objective; Dice-only ablations change only the objective.

Data	Branch	Dice only	Depth + ordinary	Depth + spatial	Depth + uniform
5% data	Plane	82.00	81.81	81.36	79.44
5% data	Unwrapped	81.68	81.89	81.47	79.72
Full data	Plane	88.29	88.19	87.94	87.93
Full data	Unwrapped	88.23	88.38	88.21	87.88

Adding depth loss to the same architecture

Data	Branch	Delta (pp)	95% case-bootstrap interval
5% data	plane position	-0.19	[-0.342, -0.044]
5% data	unfold position	+0.21	[+0.018, +0.412]
Full data	plane attention	-0.10	[-0.387, +0.188]
Full data	unfold attention	+0.15	[-0.108, +0.419]

Depth balancing slightly benefits the unwrapped branch and slightly reduces plane-branch Dice in these runs. Its added value is small relative to the 5% branch gain. Removing depth still leaves Dice-only branches far above the native Dice baseline: plane +7.83 pp and unwrapped +7.52 pp at 5%; +0.43 and +0.38 pp at full data.

Spatial gating versus the uniform control

Data	Branch	Delta (pp)	95% case-bootstrap interval
5% data	plane position	+1.92	[+1.571, +2.282]
5% data	unfold position	+1.75	[+1.540, +1.959]
Full data	plane attention	+0.01	[-0.274, +0.292]
Full data	unfold attention	+0.33	[+0.063, +0.599]

Ordinary fusion remains the best gated/ungated choice for every selected branch. Spatial selection is better than uniform attenuation at 5% and for full-data unwrapped, but it does not improve on unattenuated fusion.

Entropy is an uncalibrated proxy from a shared classifier. The uniform control matches mean gate strength only at a given model state; independently trained models can develop different entropy and residual magnitudes. These seed-0 tests cannot establish repeatability.

What changed in the errors?

Three-seed averages. FN, FP and A/P swaps are totals over 52 cases, averaged across seeds. Surface measures are case means; smaller is better.

Data	Model	Missed (FN)	Extra (FP)	A/P swaps	MSD mm	HD95 mm
5% data	Native + depth	12,992	81,426	5,317	0.999	2.288
5% data	Plane	22,053	41,973	5,147	0.736	1.773
5% data	Unwrapped	22,151	42,208	5,669	0.739	1.793
Full data	Native + depth	15,882	19,281	3,591	0.432	1.149
Full data	Plane	16,657	17,072	3,584	0.417	1.116
Full data	Unwrapped	17,317	16,359	3,488	0.414	1.115

Fewer extra voxels, with a sensitivity cost

At 5%, unwrapped versus native + depth changes the mean counts by **+9,159 missed voxels, -39,218 extra voxels and +353 A/P swaps**. Mean surface distance falls from 0.999 to 0.739 mm. Compared with plane, unwrapped has more A/P swaps in all three seeds.

At full data, the corresponding changes are **+1,435 missed voxels, -2,922 extra voxels and -102 A/P swaps**. Mean surface distance falls from 0.432 to 0.414 mm. The swap reduction is not uniform: two seeds improve, one worsens.

Unwrapped spatial gate minus ordinary fusion (seed 0)

Data	Dice pp	FN delta	FP delta	Swap delta	MSD delta mm
5% data	-0.42	-5,328	+8,902	+106	+0.022
Full data	-0.17	-1,777	+3,507	-255	+0.016

Spatial gating reduces the missed-tissue count in the unwrapped model, but adds enough extra voxels to lower Dice and worsen mean surface distance in both regimes. This is a trade-off, not an overall improvement.

What these numbers can and cannot say

FN is annotated foreground predicted as background; FP is predicted foreground outside annotation; swaps are anterior/posterior errors where both are foreground. A reduction in aggregate FN does not identify exactly which voxels were recovered, because some voxels may improve while others regress. The checkpoint audits on page 7 use paired voxel identities for that narrower comparison.

Surface metrics use the combined foreground boundary, not the anterior/posterior interface. Mean surface distance pools both directional boundary distances within a patient, then averages patients. HD95 is the analogous patient-level 95th percentile, averaged across patients.

Error distribution by depth

Absolute distance from the cutting plane, combining the two signed sides for readability. Error rate = (FN + FP + A/P swaps) / foreground union. Rates pool counts across cases and three seeds; counts in the final columns are per-seed averages.

Data	Depth mm	Native %	Plane %	Unwrap %	Unwrap FN	Unwrap FP	Unwrap swap
5% data	0-2	30.6	24.4	24.7	3,530	10,095	1,655
5% data	2-4	33.5	26.8	27.1	3,913	9,992	1,503
5% data	4-6	37.1	30.4	30.7	4,524	8,547	1,233
5% data	6-8	45.1	39.2	39.6	4,798	6,911	952
5% data	8-10	63.2	59.1	59.9	3,919	4,729	312
5% data	10+	81.3	80.9	81.3	1,466	1,933	14
Full data	0-2	14.2	13.8	13.7	2,959	3,626	1,013
Full data	2-4	15.9	15.4	15.4	3,218	3,629	916
Full data	4-6	19.2	18.8	18.9	3,675	3,483	715
Full data	6-8	25.4	24.8	24.9	3,453	2,949	576
Full data	8-10	41.3	40.6	40.4	2,701	1,972	257
Full data	10+	67.7	68.6	68.2	1,310	699	11

How to read the depth bands

The table localizes where errors accumulate; it does not establish that each band has equal anatomical difficulty. The union denominator changes with the model prediction, and sparse outer bands can have unstable percentages. Signed per-case bands and exact totals are retained in the supporting CSV/JSON.

Depth balancing remains a useful way to give sparse layers training influence. However, these results do not support the stronger claim that the added branch consistently restores missing tissue at difficult depths. The overall Dice improvement can coexist with increased FN, because FP decreases much more.

The reference plane for these evaluation bands comes from the annotated foreground. It is an evaluation/loss coordinate system. The auxiliary MRI inputs instead use the frozen guide-derived charts. A cutting-plane distance is not the distance to the outer surface and is not an SDF.

No new layer-specific or 3D signed-distance loss was tested in this follow-up. These tables should not be interpreted as evidence for such a term.

What the checkpoint audits show

These are ablations of the same jointly trained seed-0 checkpoints. Native-only here is **not** a separately trained baseline. Zero-mapped removes auxiliary features while retaining support and the learned fusion block.

Data	Branch	Native path only	Zero mapped features	Normal fusion
5% data	plane position	76.35	81.39	81.81
5% data	unfold position	78.16	81.66	81.89
Full data	plane attention	85.74	87.42	88.19
Full data	unfold attention	85.30	85.28	88.38

Mapped information contributes: removing it reduces Dice, especially for full-data unwrapped (88.38% to 85.28%). But a native-only ablation also changes the operating point of a jointly trained system, so its difference cannot isolate a standalone branch benefit.

Exact foreground changes: normal fusion versus native path

Data	Branch	Newly missed	Recovered FG	New FN >2 mm	Native entropy
5% data	plane position	13,977	0	11	0.727
5% data	unfold position	14,659	0	29	0.724
Full data	plane attention	8,120	8	0	0.553
Full data	unfold attention	9,122	0	3	0.576

Fusion primarily prunes foreground in these audits. Relative to each checkpoint's native path, it introduces thousands of missed foreground voxels and recovers zero in three conditions, only eight in full-data plane. Nearly all newly missed voxels lie within 2 mm of the annotation boundary.

This supports the concern that current fusion is an effective false-positive suppressor rather than a reliable missing-tissue repair mechanism. It does not prove that a separately trained native baseline would make exactly these same errors.

The entropy column is voxel-weighted mean normalized native entropy at newly missed foreground. It is descriptive: there is no matched correct-voxel entropy control here, so these means do not establish calibrated error detection.

Foreground recovery only means changing background to some foreground class; the A/P label may still be wrong. Surface-distance bins here use unsigned distance to the annotation surface, which differs from the cutting-plane depth bands. All normal reloads reproduced saved Dice exactly.

All results: 5% data

22 unique conditions in this regime: 10 original seed-0 models and 12 follow-up runs. Best/stop gives the selected epoch and total epochs completed. MSD and HD95 are in millimetres. FN/FP/swaps are totals over 52 validation cases.

Model / seed / loss / gate	Dice %	Best / stop	FN	FP	Swaps	MSD	HD95
Native Dice / s0	74.17	73/75	15,386	80,273	5,794	1.030	2.315
Native + depth / s0	75.12	75/75	15,273	75,585	5,461	0.959	2.182
plane local / s0 / depth	81.10	41/49	24,555	34,447	4,341	0.694	1.667
unwrap local / s0 / depth	79.81	25/33	18,832	47,451	4,774	0.762	1.752
plane context / s0 / depth	81.48	33/35	24,784	32,671	4,245	0.684	1.625
unwrap context / s0 / depth	81.19	35/36	23,855	33,402	4,957	0.682	1.622
plane position / s0 / depth	81.81	39/45	24,380	31,669	4,269	0.669	1.596
unwrap position / s0 / depth	81.89	42/50	25,800	28,661	4,562	0.658	1.603
plane attention / s0 / depth	82.06	42/50	25,271	28,929	4,456	0.651	1.565
unwrap attention / s0 / depth	81.76	42/50	25,805	29,070	4,640	0.659	1.612
plane position / s0 / Dice	82.00	39/47	23,486	31,554	4,447	0.656	1.576
unwrap position / s0 / Dice	81.68	45/52	26,649	28,237	4,597	0.663	1.606
Native + depth / s1	78.02	75/75	15,733	60,262	5,139	0.830	2.072
plane position / s1 / depth	82.10	30/38	23,032	32,330	4,521	0.656	1.629
unwrap position / s1 / depth	81.76	31/39	23,033	33,282	4,671	0.665	1.663
Native + depth / s2	71.50	74/75	7,970	108,431	5,350	1.209	2.611
plane position / s2 / depth	76.18	74/75	18,747	61,920	6,650	0.883	2.094
unwrap position / s2 / depth	75.53	74/75	17,619	64,681	7,775	0.893	2.114
plane position / s0 / depth / spatial	81.36	55/63	24,240	33,453	4,387	0.682	1.614
plane position / s0 / depth / uniform	79.44	48/50	17,962	50,604	4,776	0.780	1.775
unwrap position / s0 / depth / spatial	81.47	42/48	20,472	37,563	4,668	0.680	1.620
unwrap position / s0 / depth / uniform	79.72	48/50	18,242	48,333	4,966	0.766	1.772

Local -> context -> position -> attention are cumulative architectural variants. All original auxiliary models use depth loss. Gate variants retain depth. Full run IDs, class-specific Dice, foreground Dice, per-case metrics, epoch histories and checkpoint hashes are in all_results.csv and the evidence archive.

All results: Full data

22 unique conditions in this regime: 10 original seed-0 models and 12 follow-up runs. Best/stop gives the selected epoch and total epochs completed. MSD and HD95 are in millimetres. FN/FP/swaps are totals over 52 validation cases.

Model / seed / loss / gate	Dice %	Best / stop	FN	FP	Swaps	MSD	HD95
Native Dice / s0	87.85	23/25	17,533	17,177	3,635	0.425	1.142
Native + depth / s0	88.01	12/25	17,282	17,138	3,524	0.424	1.156
plane local / s0 / depth	88.03	12/25	16,572	17,233	3,855	0.415	1.140
unwrap local / s0 / depth	88.08	9/25	14,494	19,631	3,906	0.418	1.131
plane context / s0 / depth	88.29	12/25	16,467	17,027	3,555	0.412	1.132
unwrap context / s0 / depth	88.22	21/25	17,397	16,456	3,424	0.415	1.100
plane position / s0 / depth	88.33	9/25	15,167	19,016	3,270	0.419	1.118
unwrap position / s0 / depth	88.23	15/25	16,127	17,232	3,735	0.408	1.115
plane attention / s0 / depth	88.19	9/25	15,835	17,813	3,730	0.415	1.116
unwrap attention / s0 / depth	88.38	21/29	17,989	15,082	3,389	0.407	1.110
plane attention / s0 / Dice	88.29	11/25	15,234	18,643	3,485	0.415	1.118
unwrap attention / s0 / Dice	88.23	16/25	17,311	16,131	3,563	0.412	1.118
Native + depth / s1	87.73	13/25	14,788	20,945	3,705	0.439	1.151
plane attention / s1 / depth	88.07	14/25	16,271	18,141	3,530	0.426	1.118
unwrap attention / s1 / depth	88.13	21/29	17,258	16,821	3,473	0.418	1.127
Native + depth / s2	87.87	12/25	15,575	19,760	3,543	0.433	1.140
plane attention / s2 / depth	88.31	12/25	17,865	15,261	3,493	0.409	1.115
unwrap attention / s2 / depth	88.13	22/26	16,703	17,175	3,603	0.418	1.107
plane attention / s0 / depth / spatial	87.94	25/26	19,136	15,573	3,297	0.427	1.124
plane attention / s0 / depth / uniform	87.93	15/25	18,048	16,661	3,434	0.426	1.116
unwrap attention / s0 / depth / spatial	88.21	21/29	16,212	18,589	3,134	0.423	1.139
unwrap attention / s0 / depth / uniform	87.88	21/25	16,892	18,015	3,566	0.426	1.124

Local -> context -> position -> attention are cumulative architectural variants. All original auxiliary models use depth loss. Gate variants retain depth. Full run IDs, class-specific Dice, foreground Dice, per-case metrics, epoch histories and checkpoint hashes are in all_results.csv and the evidence archive.

Limits, preservation and cleanup

What this experiment establishes

For this fold, schedule and fixed guide geometry, auxiliary branches improve validation Dice versus the native depth model across three segmentor seeds. The matched plane branch explains most or all of that gain. Ordinary fusion outperforms the tested entropy gates at seed 0. The anatomical trade-off is fewer extra voxels with more missed tissue.

What remains unresolved

This is one reused validation fold with validation-selected architecture choices and checkpoints. It is not an independent test and does not establish clinical utility. Three seeds share the same training subsets and fixed guide pipeline. Dice-only and gating comparisons have one seed. The 75-epoch limit and early stopping describe this budget; they do not establish convergence or asymptotic rankings.

The plane control matches capacity and corresponding MRI information within a branch pair. Comparisons against the native model also change architecture, guide-derived support and contextual information, so those gains cannot be assigned to a single mechanism. Absence of a consistent unfolding gain here does not prove that every possible unfolding architecture is ineffective.

Verified evidence

Job 687700 completed successfully on 7 October 2026 at 18:38:59 Europe/Rome (exit 0), after 5 h 9 min 7 s. Verified: 24 requested follow-up completion records, 20 original conditions, four checkpoint audits, two CUDA preflights, and 1,339 recorded artifact hashes including checkpoints and geometry references.

Transferred and SHA-256 verified 792 files; 26 duplicate original files map to canonical copies. Small prediction arrays, per-case results, histories, configurations, source snapshots, guide/atlas provenance, environment, logs and scheduler records are preserved. The analysis independently recomputed every run summary, best epoch and depth-error total.

Cleanup scope

The allowlisted inventory selects 965 unshared generated files/links (4.162 GiB logical size) for deletion after this PDF and its evidence are verified. The exact paths are recorded in cleanup_plan.json.

The active tissue-profile campaign uses the original atlas and baseline result files. Those dependencies are retained, together with source code and runtime dependencies. Raw MRI/labels, unrelated jobs (including capped-depth), and local report/evidence are outside deletion scope. No active job is stopped or changed.

The final cleanup_receipt.json records deletions, errors if any, retained-reference verification and freed bytes. final_quota.json records the refreshed BeeGFS quota. These receipts are produced after report verification; the PDF is the pre-cleanup scientific record.

Evidence index

all_results.json/csv: 44 conditions. per_case_metrics.csv: 2,288 case rows. matched_comparisons.json/csv: paired differences and bootstrap intervals. seed_summary.json: seed variability. depth_band_metrics.csv and absolute_depth_summary.json: layer errors. checkpoint_audits.json: paired foreground changes. evidence/: source records. preservation_manifest.json and PDF QA receipt: local integrity and visual verification.